2023

COMPUTER SCIENCE AND ENGINEERING

Paper : CSML-1002

(Topics in DBMS)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1**, and **2** and ***any four*** questions from the rest.

1. Define and explain ***any five*** from the following : 2×5

(a) Define physical image of a global relation at a site.

(b) Define minterm predicate.

(c) State two main disadvantages of distributed databases.

(d) Define fragmentation transparency.

(e) What is the purpose of Pivot operation on a cube data model?

(f) Define data warehouse.

(g) Why newer conceptual data models are required for handling semi-structured data?

2. Comment on the correctness of the statements below and justify your opinion (***any five***) : 4×5

(a) Client applications invoked from computers at different locations, all connected to a powerful remote database server with a back-up server, is an example of a distributed database.

(b) Allocation schema for a distributed database system is location independent.

(c) Dicing does not necessarily reduce dimensions of a data cube.

(d) Concept hierarchy implies a partial order between the elements.

(e) Query processing for distributed database involves lower communication costs in general.

(f) Distributed databases offer increased availability for any database application.

(g) Online Analytic Processing (OLAP) is done on historical data only.

3. Zeus Healthcare Society (ZHS) has a chain of hospitals in four big cities across India. The information to be stored is given below:

Hospital information: hospital id, city, number of beds, bed charges.

Patient information: patient id, name, address, phone number.

Occupancy: patient id, hospital id, bed number, admission date, release date.

Design a distributed database solution that includes definition of global schema, fragmentation schema and allocation schema, for hospital bed reservation system to be accessed from multiple terminals in multiple wards located in different hospitals of ZHS considering the following frequent queries / operations :

- Request for reservation of bed : If a patient wants to book multiple beds in same or more than one hospital of ZHS, for different period, separate reservation is to be done for each hospital bed and period of booking.
- Availability of bed : The system is to check for availability of bed in a particular hospital for a period.
- Revenue query : The system is to be used to enquire about the revenue generated in a single hospital or in the entire chain for a period. State your assumptions, if any. 10

4. (a) Compare the advantages and disadvantages of centralized versus distributed database systems.

(b) Explain with an example how a single updation operation on a single data item could result in multiple updation operations on several sites when the database is physically distributed.

(c) In the context of distributed database design, why is it important to identify which part of application data are read-only items? 4+3+3

5. In an industrial production process, Robots are used to enhance production. It requires to store data pertaining to different workshops, the tools used in the workshops, parts and consumables. The tools and other components in a workshop can be described with a varied number of parameters, depending on a specific tool. The number of end-products produced in a workshop may vary and these are produced in different batch sizes.

(a) Design a semi-structured database schema for the production environment using Hyper file model.

(b) The task profiles for robots varies depending on the type of tool and workshop in which the robot is being used. We assume that a Hyper file named *RoboServices* stores the basic information like *job_ID, job_name, robo_ID. Workshop_ID.* and *cost* for different services.

- Write a query using Hyper file model to list the name and cost for all the jobs done in workshop *shop_4.*
- Write a query using Hyper file model to extract job details carried out by the robot with ID *Robo_05.* [State your assumptions, if any, clearly] 6+(2+2)

6. (a) What is constellation model? Explain with an example.

(b) A cube model has data in five (5) dimensions for a particular subject orientation. The cardinalities for the five dimensions are 3, 3, 4, 5 and 4 respectively.

(i) How many Fact and Dimension Tables are required for an equivalent Star schema?

(ii) Compute the number of tuples that would occur in each of the Dimension and Fact Tables in your design.

(c) Discuss how the structure of a cube model changes for a *Dice* operation involving selection criteria defined on two (2) different dimensions. 3+(1+3)+3

7. (a) Describe the 6-layer architecture for a distributed database system and state the roles for each layer.

(b) What is a distributed-join operation? How distributed-join operation could significantly reduce the cost of query processing in a distributed environment? 6+4

8. (a) What is semi-join operation?

(b) What is a checkpoint? How is it helpful for transaction processing?

(c) Explain the role of local mapping schema for the integration of heterogeneous multi-site database environment.

(d) Explain the roles of distributed transaction manager (DTM) and local transaction manager (LTM) towards transaction management. 1+3+3+3